

Aritra Mondal

Data Scientist | Machine Learning & Distributed Systems

Production ML (Amadeus) · Applied MSc Bac+5 (Summa Cum Laude) · Immediate Availability (CDI Cadre)

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Location: Paris & Biarritz (Relocation / Hybrid) • **Work Rights:** Titre de séjour (Full EU Rights, No Sponsorship) • **Notice:** Immediate (0-Day) — CDI Cadre

PROFESSIONAL SUMMARY

Production-focused **Data Scientist** with proven enterprise track record at **Amadeus** scaling distributed machine learning and big data pipelines for airline revenue management and demand forecasting. Specialized in **time-series predictive modeling**, high-throughput distributed compute (**Apache Spark, Scala, AWS EMR**), and automated **MLOps CI/CD**. Combines academic excellence from an **Applied MSc in Data Science & AI (Summa Cum Laude, Bac+5)** with deep computer systems fundamentals from a **BSc (Hons) Computer Security (First Class)**. Legally authorized to work across France & EU with zero notice period.

TECHNICAL COMPETENCIES

Machine Learning & AI:	**Python**, **Scikit-learn**, **XGBoost**, LightGBM, **Time-Series Forecasting**, PyTorch / LSTM, Feature Engineering, Bayesian Optimization (**Optuna**), TreeSHAP
Big Data & Data Engineering:	**Apache Spark**, **Scala**, **PySpark**, **AWS EMR**, **Parquet**, Amazon S3 Data Lakes, Advanced SQL (Window Functions, CTEs), DuckDB, Neo4j
Cloud, MLOps & Systems:	**AWS (EMR, S3, EC2, IAM)**, **Docker**, **Jenkins CI/CD**, Linux / UNIX Internals, Bash Scripting, Git Workflows, Asynchronous REST APIs
Cybersecurity & Protocols:	**TCP/IP, UDP, ICMP Telemetry**, **SNORT Network IDS**, Heuristic Inspection, Rule Optimization, Cryptography, Defensive Architecture

PROFESSIONAL EXPERIENCE

Data Scientist (Apprentice) — Amadeus

Aug 2023 – Sep 2024

Sophia Antipolis & Remote, France · Airline Revenue Management & Demand Forecasting (Order Business Domain)

- Engineered and deployed scalable production **machine learning pipelines** for passenger demand forecasting on **AWS EMR clusters** and **Amazon S3**.
- Analysed and processed enterprise **booking databases (300+ attributes, PNR records)** spanning origin/destination, layover, and timestamps with historical data back to 2010.
- Accelerated distributed batch processing performance by **35%** by migrating legacy PySpark workflows to native **Scala**, optimizing memory partitions & Catalyst execution plans.
- Constructed automated **CI/CD pipelines** using **Jenkins and Bash**, standardizing reproducible testing artifacts and reducing manual intervention by **50%**.
- Conducted functional validation, data quality investigations, and expected-vs-actual verification across **Spark and Parquet** data lakes, feeding high-throughput downstream predictive models.

Android / Software Developer (Intern) — Jungleworks

Nov 2020 – Apr 2021

100% Remote · Mobile Software Architecture & Distributed REST API Integrations

- Developed native Android client software in **Java / Kotlin**, integrating asynchronous **REST APIs** with offline SQLite caching, network resilience, and defensive error handling.
- Instituted structured unit testing suites (JUnit) and reproducible debugging workflows that eliminated core crash vectors and elevated production build stability.

FEATURED PRODUCTION & APPLIED PROJECTS

Airline Demand Forecasting Engine — Amadeus Production

Re-engineered distributed ETL & forecasting on **AWS EMR (Spark/Scala)**. Catalyst optimization delivered verified **35% latency reduction** over multi-GB Parquet datasets.

Financial Time-Series & Alpha Engine — Open Source / Quant

Multi-asset pipeline (CCXT, yfinance, MT5) with **73+ engineered alpha features**. Validated XGBoost and LSTMs with **strict purged walk-forward cross-validation (0 lookahead leakage)**.

High-Dimensional RecSys & Ranking — Applied ML

Recommendation engine on 10k+ items. Optimized via **Bayesian search (Optuna)** with **TreeSHAP feature explainability** to mitigate cold-start drop-off.

SNORT Network IDS Optimization — Cybersecurity

Real-time traffic telemetry across TCP/IP, UDP, ICMP protocols. Optimized heuristic inspection rules, reducing **false positive alarms by ~20%**.

EDUCATION & DEGREES

Applied MSc in Data Science & AI (Bac+5 Equiv.) 2022 – 2024

Data ScienceTech Institute (DSTI) · France · Summa Cum Laude / Félicitations du Jury (Rank 1)

Thesis: Production Spark/Scala pipeline optimization and airline forecasting on AWS EMR. Alternance at Amadeus.

BSc (Hons) Computer Security (Bac+3 Equiv.) 2017 – 2020

Northumbria University · UK · First Class Honours

Thesis: SNORT Network IDS heuristic inspection optimization (~20% false-positive reduction). Linux, TCP/IP, Cryptography.

Advanced Diploma in IT (Bac+2 Equiv.) 2015 – 2017

MDIS Singapore · Distinction in Information Technology · OOP (Java, C++), SQL, Distributed Systems.

CERTIFICATIONS & LANGUAGES

- **AWS Solutions Architect – Associate:** In Progress (Target Q3 2026)
- **Neo4j Certified Professional:** Neo4j Graph Academy (Verified 2024)
- **Certified Ethical Hacker (CEH):** EC-Council (Credential ECC904251, 2019)
- **Fortinet NSE 1, 2, 3:** Certified (Valid through June 2026)
- **English:** Native / Bilingual (C2 – Technical & Professional Fluency)
- **French:** Professional Working Proficiency (B1/B2, active progression)
- **Hindi & Bengali:** Native / Bilingual